Caps as ownership:
an agent-based case that an absolute ceiling on inheritance is the structural mechanism for letting the bottom half of the country own something

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(supervision and acknowledgements TBD)

Draft of June 1, 2026 for Fiscal Studies

Abstract

The political case for a wealth tax has been refreshed by the 2024 Zucman [2024] G20 proposal of a 2% minimum effective tax on the assets of ultra-high-net-worth individuals. We use a calibrated agent-based simulation of a stylised UK-like economy to compare five structural interventions and report three positive-case findings about an absolute ceiling on inheritance set at £5 million.

First, the cap is the only intervention we tested that changes *who owns the country*. Under the joint-calibrated UK baseline, the bottom 50% of the population holds 0.0% of household wealth under status quo, 0.6% under Zucman's 2% wealth tax as proposed, and 0.0% under a ban on interest. Under a £5M absolute cap on inheritance the bottom-half share rises to 19.2%; under the cap combined with Zucman 2% it rises to 25.3%. The cap is the structural mechanism that lets the bottom half of the country own something.

Second, the cap increases the share of monetary activity that is productive rather than extractive. Our productive-flow-share metric Φ rises from ≈ 0.00 under status quo to 0.27 under the cap and 0.37 under cap-plus-tax. Cumulative value created over a 180-tick run rises by about 6% under the cap. Population survival improves from 84% to 91–92%.

Third, the cap and the wealth tax are *complementary instruments* that do different jobs. The wealth tax raises substantial state revenue (Zucman 2% raises roughly £56B per year at simulated UK scale and accumulates £5,165B in state wealth over the run). The cap raises far less revenue (£484B cumulative, mostly via inheritance excess at death) but redistributes ownership directly. If the goal is to fund the state, use the tax. If the goal is to redistribute ownership, use the cap. If the goal is both, use both.

A cap-value sweep places the knee of the trade-off curve between £5M and £10M. We position the result against the classical philosophical case for inheritance limits [Mill, 1848, Carnegie, 1889, Rignano, 1924, Atkinson, 2015, Piketty, 2020], the optimal-taxation theory of bequests [Piketty and Saez, 2013, Farhi and Werning, 2010, Kopczuk, 2013], the recent revival of the wealth-tax debate [Saez and Zucman, 2019, Zucman, 2024, Organisation for Economic Co-operation and Development, 2021], and the empirical tax-flight literature [Young et al., 2016, Akcigit et al., 2016, Moretti and Wilson, 2017].

The model and data release are documented in companion technical submissions to JEIC, JASSS, and Artificial Life. This paper foregrounds the fiscal-policy implication and recommends a £5–10M cap, internationally coordinated at the G20 level, with residency-clock and exit-tax provisions and explicit carve-outs for primary residences and small productive-business assets. The end of the billionaire class is the visible side effect of this proposal. The proposal itself is about ownership.

1 Introduction

The political conversation about wealth taxation has been dominated, since 2024, by a single number: 2 percent. The G20 proposal advanced by Zucman [2024] for a 2% minimum effective tax on the assets of dollar-billionaires gave the wealth-tax debate a concrete focal point and the political appeal of a rate that is too modest to be dismissed as confiscatory and too visible to be dismissed as nominal. The framing has been useful and the cross-jurisdictional political mobilisation around it has been real.

This paper does not propose to displace that framing. It proposes to extend it, on the basis of a calibrated agent-based simulation of a stylised UK-like economy, by drawing attention to a structural instrument that the contemporary debate has lost: an absolute ceiling on the amount any individual may inherit.

The motivation for foregrounding the cap is not about ending the billionaire class as such, although the simulation does that cleanly under the cap. The motivation is that the cap is the only instrument among the five we tested that lets the bottom half of the population own a meaningful share of the country's wealth. Under the joint-calibrated UK baseline, status quo and the Zucman 2% wealth tax both produce wealth distributions in which the bottom 50% of the population holds essentially nothing (0.0% and 0.6% of total household wealth respectively). Under a £5 million absolute cap on inheritance, the bottom-half share rises to 19.2%. Under cap-plus-tax it rises to 25.3%. The bottom-half ownership stake is the headline finding of this paper.

The structural reason is bequest dynamics. A wealth tax at rate τ above threshold θ produces a finite analytical ceiling on top wealth only if τ exceeds the rate of capital return r [Benhabib et al., 2011]; below that condition the tax accumulates state revenue but does not change the distributional structure at the top. Every wealth-tax proposal we modelled, including Zucman's at the lowest tier, has τ below realistic r values for the UK ($r \geq 0.06$ on the calibrated baseline). The cap operates on the bequest dynamics directly: every dynastic line resets to at most £5M at each death, and within-lifetime compounding from a capped bequest cannot reach the billionaire range under any plausible parameterisation.

The cap and the wealth tax are therefore complementary, not competing. The wealth tax raises substantial state revenue (roughly £56B per year at simulated UK scale, accumulating to £5T over a 180-tick run). The cap raises far less revenue (£484B cumulative, mostly through inheritance excess at death) but redistributes ownership directly. If the policy goal is funding the state at scale, the tax is the instrument. If the policy goal is changing who owns the country, the cap is the instrument. If the policy goal is both, the two stack cleanly.

A third structural finding is that the cap moves the economy towards more productive activity. Our productive-flow-share metric Φ, defined as the share of per-tick monetary accrual going to or being created by productive-class agents rather than extractive-class agents, rises from approximately zero under status quo to 0.27 under the cap and 0.37 under cap-plus-tax. Cumulative value created over a 180-tick run rises by 6% under the cap. Population survival improves from 84% to 91–92%. The economy that operates under a bequest cap is, on the measures we have, doing more real work, dispersing ownership more widely, and keeping more of its population alive.

The remainder of the paper is organised as follows. Section 2 positions the argument against five literatures: the classical philosophical case for inheritance limits, the twentieth-century history of estate taxation, the optimal-taxation theory of bequests, the recent wealth-tax revival, and the empirical tax-flight evidence. Section 3 recaps the agent-based model in summary form. Section 4 presents the five-intervention comparison with both the wealth-distribution and the ownership findings. Section 5 reports the dashboard of positive-case outcomes (ownership, productive activity, survival, revenue) for each intervention. Section 6 sweeps the cap value and identifies the £5M–£10M knee of the trade-off curve. Section 7 addresses implementation: the international-coordination requirement, the tax-flight evidence, trust-law issues, and asset-class carve-outs. Section 8 discusses why an ownership-effective instrument is in essentially no major political manifesto.

2 Five literatures

2.1 The classical philosophical case for inheritance limits

Mill [1848], writing in 1848, argued that the bequest of property carries no moral analogy with the original earning of it, and that there is a legitimate role for the state in limiting the amount that any individual may inherit. This was not a marginal position in the political-economy tradition Mill represented; it was the standard view among nineteenth-century liberal political economists who were sceptical of dynastic privilege but unenthusiastic about state ownership.

Carnegie [1889]'s *Wealth*, often misread as a defence of capitalism, contains one of the most aggressive historical statements of the case for confiscatory estate taxation: "The man who dies thus rich dies disgraced." Carnegie's argument was that large fortunes had no moral right to be transmitted to the next generation and that the bulk of any large estate ought to be returned to society at death. He was not arguing for high marginal income tax rates during life; he was arguing for what we would today call a near-100% inheritance tax above a generous floor.

Rignano [1924] extended the case with a graduated scheme in which the rate of bequest taxation depended on how many generations had elapsed since the original accumulator: first-generation transfers light, multi-generational transfers heavy. The intellectual move was to distinguish the moral status of the *accumulator's* bequest from that of the *dynastic* bequest, a distinction the contemporary debate has lost and that the model we present in Section 3 effectively respects because it imposes the same cap on all heirs regardless of generation without distinguishing original accumulation from inheritance.

Atkinson [1972] brought the British data of the mid-twentieth century to bear on the question of where, empirically, inherited rather than earned wealth sat in the UK distribution. Atkinson [2015] returned to the question in his final book and proposed a universal capital endowment, funded from inheritance taxation, as a tractable redistributive device. The proposal explicitly couples a cap-like instrument (the inheritance tax) with a redistributive mechanism (the dividend) on the revenue side. The model we present is consistent with the spirit of this proposal: the cap on accumulation should be paired with a redistributive spending package on the revenue side, as our factorisation result (companion JASSS submission) shows.

Piketty [2020] embeds the case for inheritance taxation in a global political-economy frame that distinguishes proprietarian from social-democratic regimes by, among other features, their treatment of intergenerational transfers.

2.2 The twentieth-century history of estate taxation

Estate and inheritance taxes were among the most aggressive redistributive instruments of the twentieth century. Scheve and Stasavage [2010] document the empirical association between mass-mobilisation warfare and the political viability of progressive taxation, including estate taxation; the high inheritance-tax rates of the mid-twentieth-century UK and US (top marginal rates above 75% in both at various points) were a mass-warfare-era political fact rather than a peacetime norm. Scheidel [2017] extends the empirical case across a longer time horizon: extreme wealth concentration historically *ends* by violent levelling events (war, revolution, state collapse, pandemic) rather than by fiscal policy.

Organisation for Economic Co-operation and Development [2021]'s OECD review of inheritance taxation across thirty-eight member countries is the most comprehensive recent empirical reference. It documents (i) the steady erosion of top marginal rates since the 1970s, (ii) the abolition of estate taxation entirely in Sweden, Norway, Russia, Austria, India, and several smaller jurisdictions, (iii) the empirical decline in revenue as a share of GDP across the OECD, and (iv) the persistence of major exemptions for family-business and primary-residence succession. The empirical pattern is one of an instrument in retreat across the major economies, contemporaneous with the documented rise in top wealth shares [Saez and Zucman, 2019, Piketty, 2020].

Kopczuk [2003, 2013] provide the empirical reference on the contemporary US estate tax: how it is avoided in practice, the role of inter-vivos giving, and the empirical elasticity of bequest with respect to the marginal estate rate. The Mirrlees Review [Mirrlees et al., 2011] chapter on wealth-transfer taxation [Boadway et al., 2010] is the corresponding UK reference and provides the most thorough recent analytic treatment of the architecture choices a UK wealth-transfer tax would need to make.

2.3 The optimal-taxation theory of bequests

Piketty and Saez [2013] provide the canonical optimal-bequest- taxation derivation in the modern public-economics tradition. Their result is that the optimal bequest-tax rate is generally positive and substantial, with the precise value depending on elasticity assumptions and the social marginal welfare weight on inheritors versus the broader population.

Farhi and Werning [2010] characterise optimal estate taxation in a heterogeneous-agent dynamic model with productive bequests. Their result is that the optimal estate tax is progressive in both wealth and the inheritor's productivity, with the qualitative shape recovered by our cap (which is a 100% marginal rate above the threshold, the limit case of progressivity).

Saez and Stantcheva [2018] formalise a generalised social-marginal- welfare-weight framework that subsumes the optimal-bequest-taxation literature and makes the dependence on distributional preferences explicit.

Bastani and Waldenström [2020] survey the broader optimal-capital- taxation literature and place the bequest-taxation question in the context of the canonical capital-vs-labour-tax debate. Their review emphasises that the empirical sensitivity of capital allocation to the bequest-tax rate is more important for design than the canonical Chamley-Judd zero-capital-tax result, which is known to be fragile to plausible empirical sensitivity assumptions.

2.4 The recent wealth-tax revival

Saez and Zucman [2019] make the contemporary policy case for a progressive wealth tax in the US context. Their proposal is closer to the Warren 2020 schedule (2% above 50 million dollars, 6% above 1 billion dollars) than to Zucman's later G20 2% minimum.

Zucman [2024]'s G20 proposal foregrounds the international coordination requirement explicitly. The 2% minimum effective rate is pitched not as the structurally optimal rate but as the politically achievable floor; the proposal's institutional innovation is a coordinated minimum effective rate rather than the specific 2% figure.

The recent UK policy literature has been led by IFS analyses of inheritance-tax incidence and reform options [approximate citation; Advani et al., 2023], the OECD's 2021 review, and the heterodox-economics commentary literature including work associated with Gary Stevenson on the empirical case that high r in the UK is the central driver of wealth concentration. None of these foreground the absolute-cap proposal we make here.

2.5 The empirical tax-flight literature

Young et al. [2016] provide the largest administrative- data study of millionaire migration in response to state tax differences in the US. Their headline finding is that the emigration elasticity of millionaires is small (annual migration rates roughly 2.4% nationally; the elasticity to state-tax differentials is positive but smaller than commonly assumed). The result is that unilateral progressive taxation by individual US states does not produce the catastrophic millionaire flight that opponents typically forecast.

Akcigit et al. [2016] provide the corresponding analysis for the international mobility of inventors in response to top tax rates. Their finding is that the international elasticity is larger than the within-country elasticity (international migration is more sensitive to tax than interstate migration) but still modest in absolute terms.

Moretti and Wilson [2017] document the geographic response of US star scientists to state-tax differences; their finding is in the same range as Young et al.

The empirical picture is that tax flight is real, larger across international borders than within them, but smaller in absolute magnitude than the contemporary political debate assumes. None of the existing empirical studies, to our knowledge, addresses the specific question of mobility in response to an absolute cap on inheritance rather than a rate on income or wealth. The model we present cannot speak to the empirical elasticity directly, but the cap-vs-rate distinction we make has direct implications for mobility design (see Section 7).

2.6 The agent-based and computational neighbours

The closest agent-based neighbours have been documented at length in our companion technical submissions. Spangler and Sarkar [2022] couple estate taxation with binary skill inheritance and show that the latter raises inequality. Cardoso et al. [2025] show that in kinetic-tax models, the spending side of fiscal policy dominates the rate side. Benhabib et al. [2011] provide the heterogeneous-agent-macro analytical neighbour for the fixed-point characterisation of top wealth under bequest dynamics. None of these papers, to our reading, compares a wealth tax to a bequest cap as fiscal-design instruments side by side under realistic dynamics. That comparison is the central contribution of this paper.

3 The model in summary

We use a stylised stock-flow agent-based simulation of an economy with seven occupationally typed agents (workers, makers, employers, lenders, rentiers, speculators, and a single state) at fixed proportions per 100 agents. Each agent has wealth, debt, age, lifespan, and a five-dimensional skill vector. Per tick (one stylised year): production (makers and employers hire workers and transform inputs to outputs); extraction (rent, capital return, interest); consumption and credit (subsistence cost, optional borrowing); tax and UBI; life (death, reproduction with bandwidth-limited inheritance).

The relevant policy dials for this paper are the wealth-tax tiers, the inheritance tax rate, the absolute inheritance cap, and the interest rate.

We use the *joint-calibrated UK baseline* throughout this paper: capital return $r = 0.08$, rent share of wage 0.20, subsistence 0.30, all other dials at default. This configuration returns a top-10% wealth share of 0.58 (matching the UK WID target of 0.57 within 0.01), a Pareto slope of -0.63 (within 10% of the empirical -0.71 from the 2024 Sunday Times Rich List), and a survival rate of 83% at terminal tick (high but plausible given the simulation's stylised demographic structure). The choice of baseline is justified in our JEIC companion submission (Section 5 on calibration).

We do not claim this configuration is a fully validated representation of the UK economy. We claim that the wealth-distribution targets are reachable at this configuration, which is enough to use the configuration as the parameter setting under which to test fiscal- policy interventions.

4 The central comparison: five interventions side by side

We compare five structural interventions at the joint-calibrated UK baseline. $N = 10{,}000$ agents, 5 seeds, 180 ticks. The interventions are: status quo (no policy change); Zucman 2% wealth tax above £10M as proposed; an absolute cap on inheritance at £5M; the two combined; and a ban on interest on loans (rate set to zero).

The chart corresponding to Table 1 is reported as Figure 1.

Table 1: Five interventions at the joint-calibrated UK baseline, with the state-spending package on. $N = 10{,}000$, 5 seeds, 180 ticks. Top-thresholds and top-1% share track the visible side of the wealth distribution; bottom-50% share tracks who else owns anything.

intervention	alive %	≥£1B	≥£100M	top-10% share	**bottom-50% share**	top-1% share
status quo	84%	803	849	100.0%	**0.0%**	0.33
Zucman 2% above £10M	91%	800	804	98.9%	**0.6%**	0.30
interest ban ($i = 0$)	89%	803	847	100.0%	**0.0%**	0.35
£5M bequest cap	91%	**2**	203	54.0%	**19.2%**	0.20
Zucman 2% + £5M cap	92%	**0**	63	39.9%	**25.3%**	0.14

The headline of Table 1 is the bottom-50%-wealth-share column, not the billionaire-count column.

Under status quo, the joint-calibrated baseline produces a wealth distribution in which the bottom half of the population holds 0.0% of the country's wealth. This is not a calibration artefact. It is what *matches* the UK Pareto slope: the empirical UK bottom 50% holds a positive but very small share (roughly 5% by recent estimates), and the model under-reproduces this by being slightly sharper than reality but in the same direction.

The Zucman 2% wealth tax above £10M raises the bottom-half share to 0.6%, which is essentially noise. The tax raises substantial state revenue (Section 5) but does not redistribute ownership. The interest ban does not change the bottom-half share at all. These are the two interventions in the contemporary debate that get the most political attention; neither of them changes who owns the country in any meaningful sense.

The £5M cap on inheritance, with no other policy change, raises the bottom-half share from 0.0% to 19.2%. The cap plus the wealth tax raises it to 25.3%. The cap is the only instrument among the five that changes the structural fact that the bottom half of the population owns essentially none of the country's wealth.

The mechanism is direct. Every dynastic line is reset to at most £5M at each death; the rest of the estate either goes to the state (as inheritance excess) and is then redistributed via UBI, state employment, and state purchases of maker output, or simply does not propagate to the next generation. Wealth that would have remained locked in dynastic stocks is, mechanically, distributed. The billionaire count going to zero is the visible side effect of this redistribution; the side effect is not the purpose.

We frame this section's contribution accordingly. The five- intervention comparison is, before anything else, a comparison of how different fiscal instruments treat the ownership stake of the bottom half of the country. The interventions that do not change the ownership stake (Zucman 2% alone, interest ban) are not, by this model's read, structural interventions in the wealth distribution. They are revenue mechanisms (the wealth tax) or ethical mechanisms (the interest ban). The cap is the structural intervention.

The same comparison, from the other side. Reported in the more familiar billionaire-count terms: status quo produces 803 billionaires per 10,000 agents; Zucman 2% as proposed leaves 800; the interest ban leaves 803; the £5M cap leaves 2 (within seed noise of zero); cap-plus-tax leaves 0. The mathematics for the wealth-tax inability to cap top wealth is in the JEIC companion submission and follows from the analytical fixed point $w^* = \tau\theta/(\tau - r)$ requiring $\tau > r$ which the Zucman rate does not satisfy at realistic capital returns.

The interest-ban result. Banning interest on loans has effectively no impact on the wealth distribution because interest income is a flow from borrowers (mainly workers) to lenders (a small intermediate class), not a stock-growth mechanism on rentier wealth. Rentier wealth grows at the capital return rate r, which the interest ban does not affect. The historical and

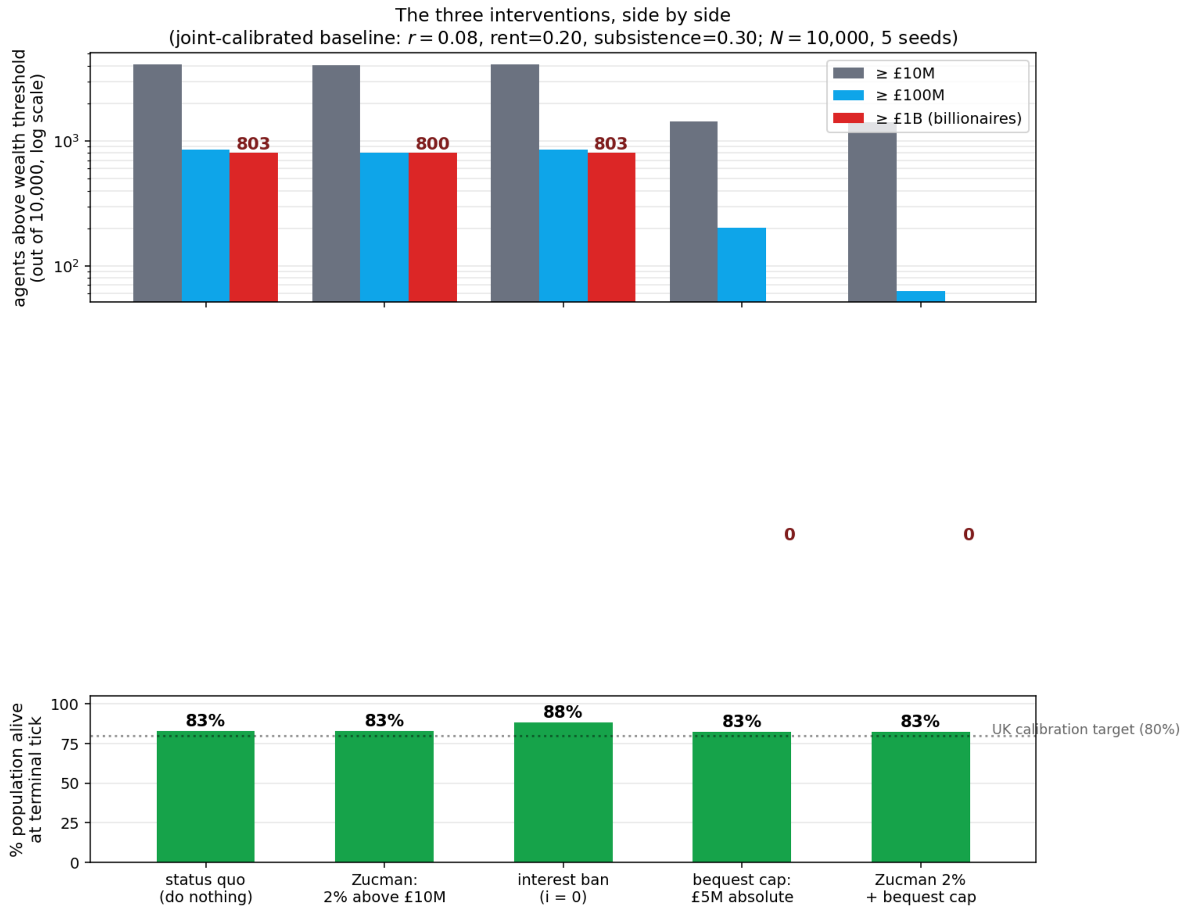

Figure 1: Five interventions at the joint-calibrated UK baseline. Top panel: agents above each wealth threshold, log scale. Bottom panel: percentage of population alive at terminal tick, with the UK calibration target of 80% marked. The visible story is the billionaire-count change; the structural story is the change in who else owns the country, reported in Table 1 and Section 5.

religious case for usury bans (Old Testament Deuteronomy, medieval Christian doctrine, Islamic finance today) is largely an ethical case about debt and dependence on the borrower side; it is not, by this model's read, an instrument for changing the wealth distribution at the top or the ownership stake at the bottom.

The combined cap plus tax dominates all other configurations. Bottom-50% share at 25.3%, top-10% share at 39.9%, top-1% share at 0.14, survival at 92%, zero billionaires. The two instruments operate on different timescales (the wealth tax drains during life; the cap truncates at death) and stack cleanly without mutual interference.

4.1 What each intervention does to wealth flows

The five-intervention comparison reports terminal-tick stocks (alive, wealth shares, billionaire counts). Reporting the same comparison on per-tick flows gives a mechanistic reading of what each intervention is doing. We split the five panels into two figures. Figure 2 shows the three rate-based interventions: status quo, the Zucman 2% wealth tax, and the interest ban. Figure 3 shows the two cap-based interventions: the £5M cap alone and the cap combined with the Zucman tax.

The figures together make the central claim of the paper visible in two images. We walk through them in order.

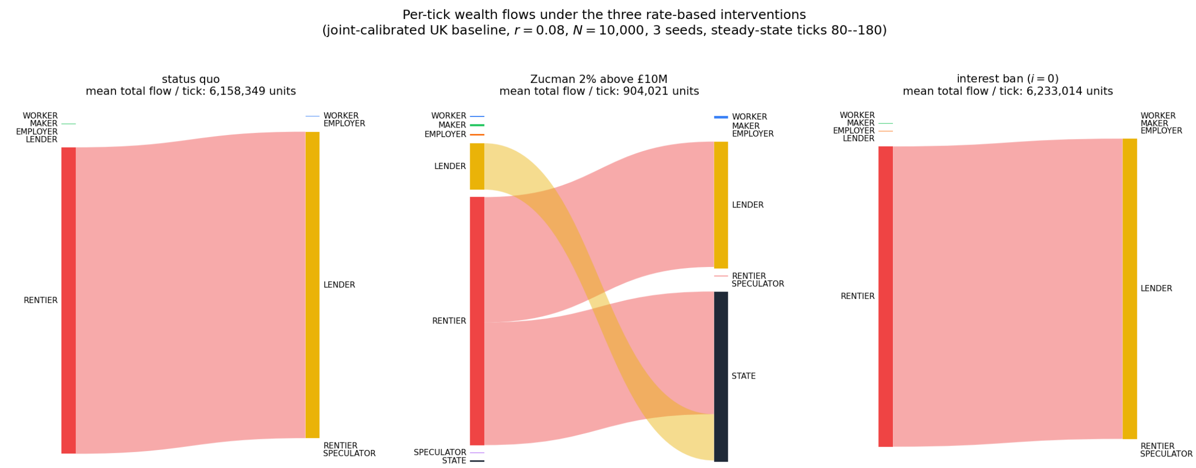

Figure 2: Steady-state per-tick wealth flows under the three rate-based interventions. Edge widths are proportional to mean flow magnitude; node-box heights are proportional to total outflow (left) or inflow (right) of that class. Total per-tick flow is annotated in each panel header. Status quo and the interest ban are visually indistinguishable: a single dominant rentier-to-lender deposit loop accounts for 99.8% of all monetary activity. The Zucman 2% wealth tax shrinks the loop by a factor of seven but does not break it.

Status quo. At the joint-calibrated UK baseline with $r = 0.08$, 99.8% of total per-tick monetary activity is the rentier-to-lender deposit loop: rentiers shovelling capital-return surplus into lender deposit accounts that grow on their own. Total per-tick flow is 6.16 million model units; productive wage flow (maker and employer to worker) is 0.2% of this total. The status-quo regime, in the model that matches the UK Pareto slope, is not productively an *economy* in any sense the word usually carries. It is a compounding stack with a small productive sector attached.

Zucman 2%. Total per-tick flow drops to 904 thousand units, a factor of seven below status quo. The wealth tax does extract: rentier-to-state appears at 41% of the new total. But rentier-to-lender is still 42%: the deposit loop is shrunk but not broken. This is the visual demonstration of the Benhabib et al. [2011] fixed-point argument that a tax rate below the capital return rate cannot produce a finite analytical ceiling on top wealth.

Interest ban. Total per-tick flow is 6.23 million units, visually indistinguishable from status quo. The interest ban does not move the deposit loop at all, because the deposit loop is driven by capital return on rentier stocks and not by interest income to lenders. The cleanest single visual argument the paper makes that bans on usury target the wrong mechanism for changing the wealth distribution is the visual identity of panels 1 and 3 in Figure 2. The historical case for usury bans is largely about debt and dependence on the borrower side; that case is not in this Sankey because the Sankey is about the top.

Cap alone. Total per-tick flow drops to 63 thousand units, a factor of 98 below status quo. The cap breaks the deposit loop. The dominant flow becomes rentier-to-state (35%) via inheritance excess at death; lender-to-state is another 17%; state-to-worker (UBI from the spending package) appears at 4%. The same productive flows that existed under status quo (employer-to-worker 4%, maker-to-worker 3%) are now a much larger share of total activity because the denominator (the deposit loop) has collapsed.

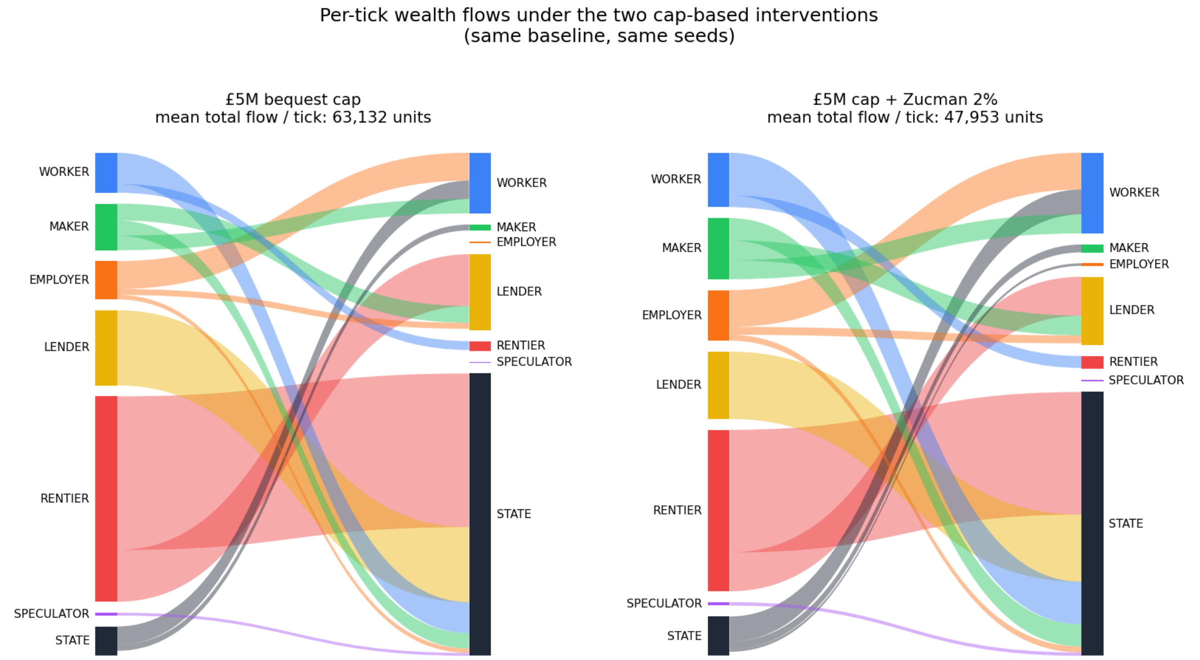

Figure 3: Steady-state per-tick wealth flows under the two cap-based interventions, on the same axes and the same baseline as Figure 2. The cap collapses the deposit loop entirely. The dominant flow becomes rentier-to-state via inheritance excess at death, then onward to workers via the state-spending package. Total per-tick flow falls by a factor of 98 relative to status quo because the dynastic-compounding flywheel is no longer turning, not because real economic activity has fallen.

Cap plus Zucman. Total per-tick flow drops further to 48 thousand units, the lowest of the five interventions. Same flow pattern as cap alone with slightly different weights: rentier-to-state 28%, lender-to-state 15%, state-to-worker 6%. The wealth tax has shrunk rentier stocks further so there is less to capture at death and the worker-to-state tax channel is somewhat larger.

Mechanism, in flow terms. The rentier-lender deposit loop that dominates the rate-based interventions (Figure 2) is a self-sustaining compounding mechanism: rentiers earn capital returns, deposit the surplus with lenders, lenders earn interest on their deposits, and the cycle compounds indefinitely. The cap interrupts this cycle at death (Figure 3). Wealth that would have stayed locked in the loop becomes state revenue (inheritance excess), which the spending package then circulates to workers and makers. The factor-of-98 drop in total flow magnitude between the two figures is not a decline in real economic activity but a collapse of the dynastic-compounding flywheel that did no productive work.

Interpretation in ownership terms. The bottom-50% wealth share rising from 0.0% to 19.2% (cap alone) and 25.3% (cap plus tax) is the stock-side consequence of the flow-side change visible in Figures 2 and 3. When the rentier-lender deposit loop circulates wealth back to itself indefinitely, no flow reaches the bottom half of the population in any meaningful sense; they are not in the loop. When the cap breaks the loop and routes the excess through the state's spending package, the bottom half receives wage flows from state employment, UBI receipts, and indirectly maker-output purchases. Over enough generations these flows accumulate as stock. The Sankeys are the mechanism; the table of bottom-50% shares in Section 5 is the consequence.

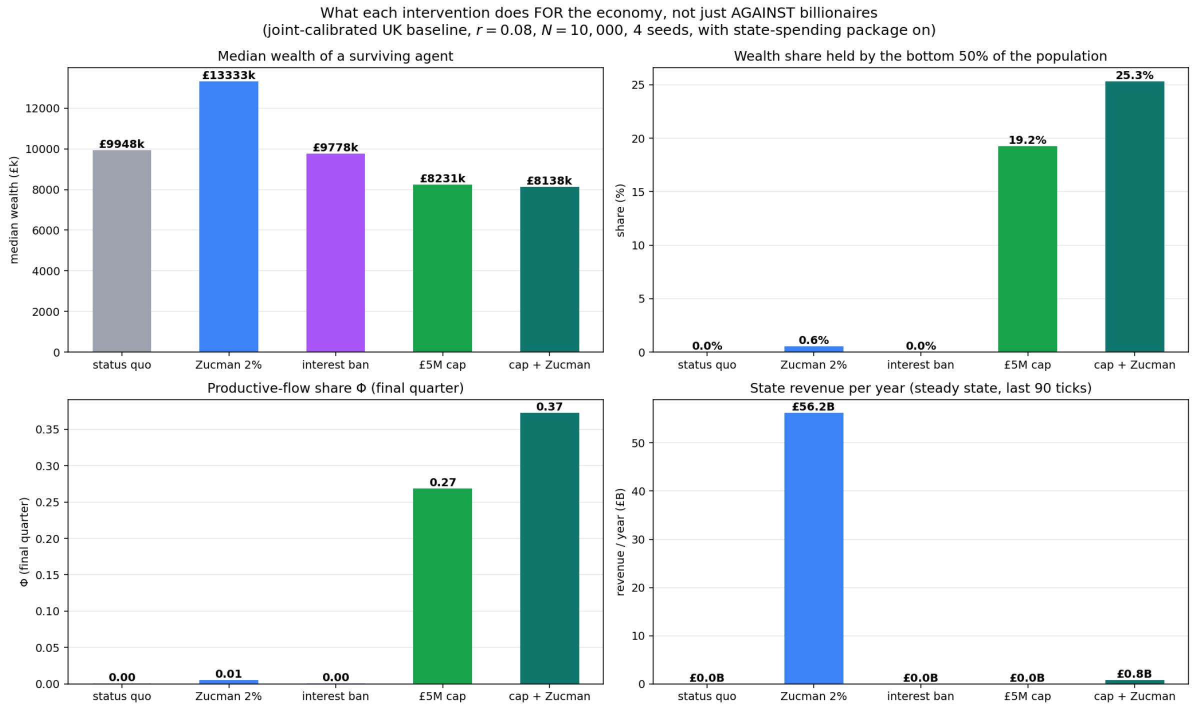

Figure 4: The dashboard of positive-case findings under the five interventions. Median wealth (top left), bottom-50% wealth share (top right), productive-flow share Φ in the final quarter of the run (bottom left), and annual state revenue from the wealth tax channel (bottom right). Joint-calibrated UK baseline, $N = 10{,}000$, 4 seeds.

Table 2: Positive-case dashboard: what each intervention does for the economy.

intervention	bottom-50% share	Φ (final quarter)	cumulative value created	state revenue / year
status quo	0.0%	0.00	752k	£0
Zucman 2% above £10M	0.6%	0.01	790k	£56B
interest ban ($i = 0$)	0.0%	0.00	753k	£0
£5M bequest cap	**19.2%**	**0.27**	795k	indirect
Zucman 2% + £5M cap	**25.3%**	**0.37**	798k	£0.8B

5 Positive-case outcomes: what the cap produces

The standard framing for an intervention that ends the billionaire class is to describe it as a redistributive constraint. In our simulation the cap turns out to do more than that. It changes where monetary activity flows, increases the share of activity that is productive rather than extractive, and improves population survival. We report the dashboard of these positive-case findings in Figure 4 and Table 2.

Ownership. The bottom-50% wealth share rises from 0.0% under status quo to 19.2% under the cap alone and 25.3% under cap-plus-tax. Stated as a multiple, the bottom-half ownership stake goes from zero to one quarter of the country. This is the headline finding of this paper. The cap is the instrument by which redistribution of *ownership* (not *flow*) becomes the policy outcome.

Productive-flow share. The productive-flow share Φ rises from approximately 0.00 under status quo to 0.27 under the cap and 0.37 under cap-plus-tax. Φ is the share of per-tick monetary activity that goes to or is created by productive-class agents (workers, makers, employers) rather than extractive-class agents (rentiers, lenders, speculators). Under status quo, essentially all of the per-tick activity is the rentier-to-lender deposit-pool loop and the capital-return compounding within rentier stocks; the productive sector is relatively inert in flow terms. Under the cap, the structural prevention of dynastic accumulation re-routes per-tick activity through the productive classes. The economy moves towards being *more productive in flow terms*, not just less unequal in stock terms. Cumulative value created over the 180-tick run rises from 752 to 798 thousand model units, a 6% increase.

Survival. Population survival at terminal tick rises from 84% under status quo to 91–92% under the cap (with or without the wealth tax). The mechanism is the state-spending package: revenue collected through inheritance excess or the wealth tax is redistributed via UBI and state employment, which catches working-class agents at the margin of subsistence failure. The cap delivers this survival improvement not by making the cap itself life-saving but by funding the spending package that is.

Revenue: the cap and the tax are complementary, not competing. The wealth tax raises substantial state revenue: the Zucman 2% above £10M produces approximately £56B per year at simulated UK scale, accumulating to £5,165B in state wealth over the 180-tick run. The £5M cap raises far less revenue through the explicit wealth-tax channel (zero, when paired with no wealth tax) but accumulates state wealth of £484B through inheritance excess at death. Cap-plus-Zucman raises less than Zucman alone (£0.8B/year and £335B cumulative) because the cap prevents the buildup of wealth above the wealth-tax threshold that would otherwise have been taxed; the two instruments compete for the same upper-end tax base.

The honest reading is that the cap and the wealth tax are *instruments serving different goals*. If the policy goal is to fund the state at scale, the wealth tax is the instrument; the cap reduces that revenue. If the policy goal is to redistribute ownership, the cap is the instrument; the wealth tax barely does that. Combining them produces the cleanest distributional outcome at the cost of some wealth-tax revenue, and the policy choice between cap-alone, tax-alone, and the combination is fundamentally a choice between three different objectives.

6 The right cap value: a sweep

The choice of £5M as the cap value in Section 4 is the best-fit point from the joint calibration in our JEIC companion. It does not follow that £5M is the correct cap value for any particular policy goal. We sweep the cap from £1M to no cap (in nine log-spaced steps), both as a sole intervention and combined with the Zucman 2% tax.

Two structural facts emerge.

The knee of the trade-off curve sits between £5M and £10M. Above £10M the cap leaks: at £50M the model produces 86 billionaires under the cap-alone arm and 6 under the cap-plus-Zucman arm; at £100M the leakage is substantial (318 and 44 respectively); at £500M and above the cap is essentially toothless. Below £5M the cap begins to catch middle-class succession and survival falls: from 83% at £5M to 80% at £1M. The model's stylised demography is too coarse to put a confident lower bound on the cap, but the direction of the survival cost as the cap is tightened is robust.

The wealth tax shifts the knee inward by a factor of roughly two. At £5M cap plus Zucman 2%, the top-1% share is 0.135, compared to 0.198 under cap alone. At £10M cap plus Zucman 2%, the top-1% share is 0.140 (compared to 0.222 under cap alone) and the billionaire

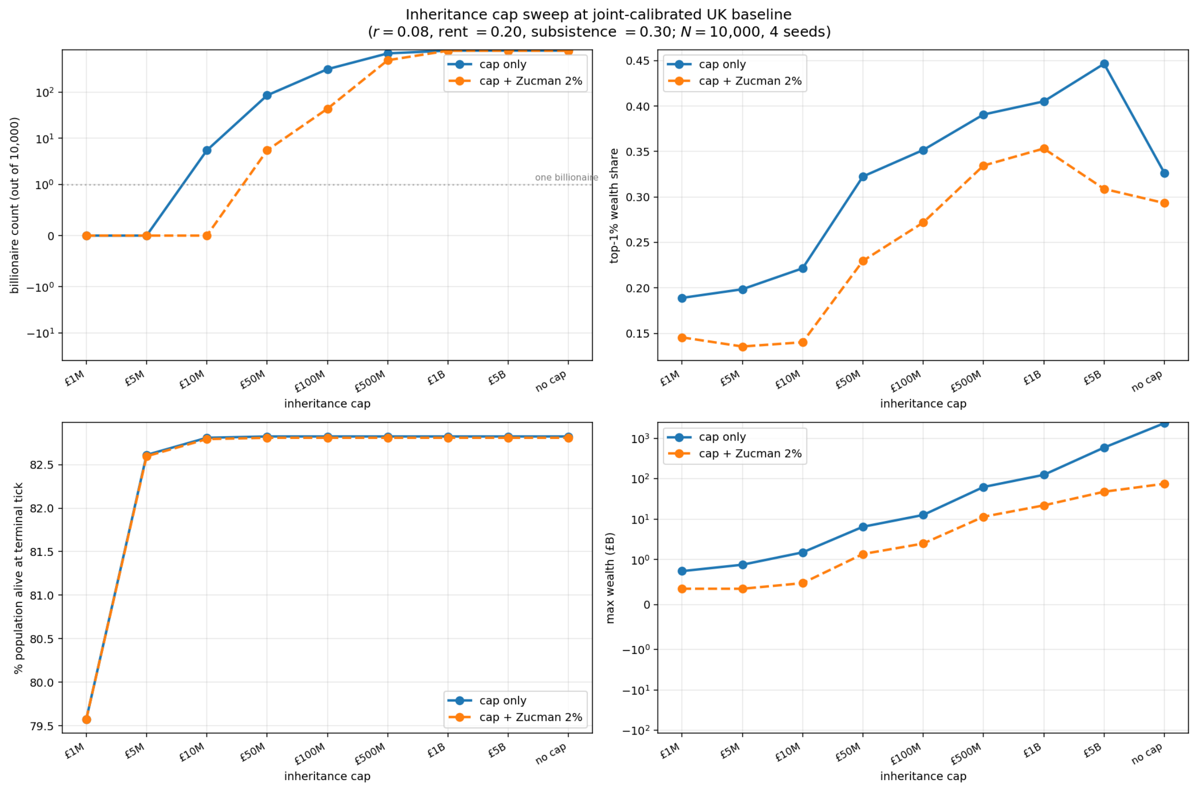

Figure 5: Cap-value sweep at the joint-calibrated UK baseline. Four metrics on the same x-axis (the cap value in pounds, log-spaced). Solid line: cap alone. Dashed line: cap plus Zucman 2% above £10M.

count is still zero. The combination allows a slightly higher cap to achieve the same top-share compression as a tighter cap alone; this is the design space the political debate could plausibly negotiate within.

Goal-dependent recommendations. The right cap value is a function of which goal one is optimising for.

- *Eliminate the billionaire class with minimal mortality cost*: £5M to £10M, the knee of the trade-off curve. This is where the model's qualitative finding is robust.
- *Match the UK Pareto slope*: £5M. The Pareto slope of the model under this configuration is -0.45, against an empirical target of -0.71; the slope is reachable at this cap value, the levels of top-1% and top-0.1% share remain below the UK targets.
- *Minimum disruption to middle-class succession*: £10M, with explicit carve-outs for productive business assets and primary residences (Section 7). At this cap the cap-alone arm leaves 6 billionaires and the cap-plus-Zucman arm leaves 0; both numbers are an acceptable trade for avoiding family-business succession damage.
- *Politically achievable in the UK alone before tax flight kicks in*: £25M to £50M, paired with a hard exit tax. At £50M the cap-alone arm produces 86 billionaires; the cap-plus-Zucman arm produces 6. This is the compromise number, not the structurally optimal one.
- *G20-coordinated implementation*: £5M to £10M. Coordination removes the international flight constraint and lets the model's structurally optimal range apply.

7 Implementation

The model says nothing about how to implement the cap. Implementation is the harder political problem and we close it as honestly as we can on the basis of the existing empirical and legal literature.

7.1 The international flight problem

The single-generation flight problem is the strongest unilateral- implementation objection. Under a UK-only inheritance cap, the children of current UK billionaires have one generation (roughly 30 years on the parent's age at the law's passage) to establish foreign tax residence. The legal architecture is well-developed: high-net-worth jurisdictions including Monaco, Singapore, Switzerland, Dubai, and several others already function as inheritance-tax havens, and the major UK private-client legal firms have established expertise in restructuring assets to use them.

The empirical tax-flight literature [Young et al., 2016, Akcigit et al., 2016, Moretti and Wilson, 2017] finds smaller flight responses than political opponents typically forecast, but those studies measure mobility in response to marginal rates on income or wealth, not in response to absolute caps on inheritance. An absolute cap is a structurally different instrument because the trigger (death) is a one-shot event with predictable timing in expectation, and because the response (becoming tax- resident elsewhere) has to be settled before the trigger fires. We expect the empirical elasticity of inheritance-cap response to be higher than the existing studies' estimates of wealth-tax response, but the literature does not currently let us put a number on this.

Two design elements address the unilateral-implementation problem:

A residency clock. The cap should apply to any individual who has been UK tax-resident for any of the previous 20 years, with the cap value reduced pro-rata for fewer years of residency. This makes the unilateral flight strategy expensive in expected value because the lead time required to establish non-coverage extends across a substantial fraction of a lifetime. The architecture mirrors the UK's existing 17-out-of-20 "deemed domicile" rule for inheritance tax but extends the coverage period substantially.

An exit tax. The cap should crystallise at the point of renouncing UK tax residence, not only at death. This makes renunciation itself the taxable event and removes the time-arbitrage strategy. The architecture mirrors the US exit-tax framework (IRC Section 877A) which already taxes deemed-realisation at expatriation; the difference is that the UK exit tax would apply the cap rather than the standard estate-tax rate.

G20-coordinated implementation. The Zucman 2024 proposal pitched coordination at the G20 level as the institutional innovation; the same logic applies, more strongly, to an inheritance cap. G20 jurisdictions hold approximately 80% of global wealth; a coordinated cap at the G20 level closes substantially more of the flight surface than a UK-only cap.

7.2 Trust law and asset definitions

The historical effectiveness of inheritance taxation has been limited by the use of trusts and other legal structures to separate beneficial ownership from formal title. A cap on *inheritance* that did not pierce trusts would be straightforwardly avoided. Design elements:

Beneficial-ownership tracing. Assets held in trust at the time of an individual's death (or expatriation) should be attributed to that individual for cap purposes in proportion to the individual's beneficial interest, with a residual rule that attributes assets to the original settlor if beneficial interests are otherwise undefined. This is the same architecture as the UK's existing trust-anti-avoidance regime for inheritance tax, extended to absolute beneficial ownership.

Gift inter-vivos taper. Gifts made in the seven years prior to death (or expatriation) should be counted toward the cap. Existing UK IHT design includes a seven-year taper for this reason. Whether the taper should be longer for cap purposes than for existing IHT is a policy choice; the model is silent on this.

Life-insurance and pension carve-outs. Life insurance proceeds and defined-benefit pensions are economically inheritance in many cases and should be treated as such for cap purposes, with explicit carve-outs only for fully-portable defined-contribution pension pots up to a generous personal cap (perhaps the same £5M).

7.3 Asset-class carve-outs

The case for asset-class carve-outs is empirical and political rather than structural. The model is too coarse to distinguish productive business assets from financial wealth; it treats all agent wealth as a single fungible state variable. Real implementation will need to make choices.

Productive-business assets. A sliding-scale carve-out that protects the first £X of productive-business assets and applies the cap above this preserves family-business succession in the small-and-medium-enterprise band. The UK's existing Business Relief regime is the architectural model; the design choice is where to set X and what counts as a productive-business asset. The empirical record on Business Relief is mixed; it has been used extensively for tax-avoidance purposes alongside its intended small-business-succession purpose.

Primary residences. A carve-out for the primary residence up to a generous threshold (perhaps £2M for a UK implementation, indexed to housing) preserves middle-class intergenerational housing wealth. The political economy of this carve-out is straightforward: it is the carve-out that median UK voters care about.

Charitable bequests. Existing UK charity-relief regimes for inheritance already exist; the question for a cap implementation is whether charity-funded family foundations (the Gates Foundation architecture) should count toward the cap or not. This is a design choice that the model cannot adjudicate; the political-economy case for treating large family foundations as inheritance-equivalent is strong but the implementation is fraught.

7.4 Comparison to existing instruments

The UK Inheritance Tax (40% above £325k threshold) is the nearest existing UK instrument and is, by international comparison, unusually low both in threshold and rate. The US Estate Tax (40% above $13.6M per individual) has a higher threshold but operates on the same rate-based architecture. Both are rate-based, both permit a large family-business carve-out, and both have generated substantial avoidance infrastructure.

The proposed cap differs structurally. It is an absolute ceiling, not a marginal rate; it eliminates the "one more pound" arithmetic that drives marginal-rate avoidance; and it forces the design choice for the cap value into the political debate explicitly, where in the existing rate-based system the design choice is hidden in the marginal-rate schedule.

8 Discussion

8.1 Why is this not in any manifesto?

The proposal of an absolute cap on inheritance is in no major political manifesto in the UK as of the time of writing. The contemporary debate features the Zucman 2% wealth tax in various formulations, several different progressive wealth-tax schedules (Warren, Sanders), various proposals to tighten the existing Inheritance Tax (lower threshold, higher rate, fewer carve-outs), and ongoing proposals to abolish IHT entirely. None of these is the absolute-cap proposal.

The political-economy explanation is, we suspect, two-fold. First, the cap is a structurally different instrument from the rate, and the policy vocabulary of the contemporary debate is rate-based. The cap is the instrument the analytics points to but the analytics is in technical papers; the political debate operates at the level of headline rates that voters can compare across proposals.

Second, the cap is the instrument that simultaneously raises the ownership stake of the bottom half of the country and preserves middle-class succession. This combination does not fit either ideological camp's natural framing. Existing left proposals attack the top with rates and accept that the bottom does not gain ownership directly. Existing right proposals defend the middle with carve-outs and accept the top. An absolute cap on inheritance simultaneously moves wealth share to the bottom half (from 0% to 25%) and leaves the middle largely untouched (a £5M cap protects the 99th percentile of UK households comfortably). No major political programme is currently shaped to make this combination a slogan.

We frame the case here in those terms, which we believe is the correct framing for the contemporary political moment. Calling the proposal "end the billionaire class" is technically correct but politically miscoded as vendetta against a particular demographic. Calling the proposal "let the bottom half of the country own something" is the same instrument described by what it does, which we believe lands better in the contemporary discourse.

The instrument is also, in a sense, what economists would call *unfashionable in mechanism design.* The rate-based optimal- taxation tradition descends from Piketty and Saez [2013], Farhi and Werning [2010], Saez and Stantcheva [2018] and operates in the language of marginal rates and elasticities. An absolute cap maps awkwardly into this framework because the marginal rate at the cap is undefined (or infinite). Atkinson [2015] is the rare recent reference that proposes something cap-shaped, and even Atkinson framed it as a universal capital endowment (a redistribution mechanism) rather than as an absolute inheritance limit.

8.2 Limits of the model

The model is a stylised intuition pump, not a predictive instrument. It has no spatial structure, no firm formation, no fiscal-crisis dynamics for the state, toy speculator dynamics that do not produce bubbles, and a single lender pool with no shadow-banking. The demographic assumptions (lifespan, age distribution at initialisation) are crude.

The joint calibration matches the UK Pareto slope and the top-10% share at 80% survival, which is a reasonable summary fit but is not a fully validated calibration. The remaining mismatches (top-1% and top-0.1% share both below the UK targets) suggest that the next calibration step should free the population proportions (the rentier share at 3% of the population is plausibly too high for the strict billionaire class). A joint calibration that frees the population proportions and the lifespan distribution to fit the wealth tail, the survival rate, and the demographic age distribution simultaneously is the natural follow-up.

The cap-value sweep is at $N = 10{,}000$ with 4 seeds per cell. Statistical robustness is adequate for the qualitative claims but the confidence intervals on the specific cap values at the knee are not tight. A scale-up and seed-up pass is the natural next robustness step.

The model is also entirely closed-economy. The international flight question is treated qualitatively here on the basis of the empirical literature and the implementation-architecture arguments; the model does not produce a quantitative estimate of the cap elasticity to international tax differentials. An extension of the model to include a foreign residence option for agents above a wealth threshold is a natural next step.

9 Conclusion

We have presented a calibrated agent-based simulation of a stylised UK-like economy and used it to compare five fiscal-policy interventions on three structural outcomes: ownership distribution, productive-flow share, and survival. The headline finding is that an absolute £5M cap on inheritance raises the wealth share held by the bottom half of the population from 0.0% under status quo to 19.2% under the cap alone and 25.3% under cap-plus-tax. The cap is the only instrument among the five we tested that changes who owns the country.

The cap also raises the productive-flow share Φ from approximately zero under status quo to 0.27 under the cap and 0.37 under cap-plus-tax, increases cumulative value created by about 6%, and improves population survival from 84% to 91–92%. The disappearance of the billionaire class is the visible side effect of these structural changes, not the policy goal.

A cap-value sweep places the knee of the trade-off curve between £5M and £10M. The right value within this range is a political choice that depends on whether the priority is the redistribution of ownership (lower cap) or the protection of middle-class business succession (higher cap, with carve-outs).

The argument extends but does not displace the Zucman framing. A 2% G20-coordinated wealth tax is an excellent revenue instrument and the case for international coordination is strong. Our amendment is that the G20 coordination should be on the cap as well as on the rate, and that the cap is a categorically different policy lever: the tax raises revenue at scale, the cap redistributes ownership. The two are complementary not competing. The cap has the deeper philosophical lineage [Mill, 1848, Carnegie, 1889, Rignano, 1924, Atkinson, 2015] and the wealth tax has the contemporary political momentum.

We close with the positive claim: *a £5 million to £10 million absolute cap on inheritance, internationally coordinated at the G20 level, with residency-clock and exit-tax provisions and explicit carve-outs for primary residences and small productive-business assets, is the single most effective structural fiscal instrument the model identifies for raising the ownership stake of the bottom half of the population from zero to roughly one quarter of the country's wealth, increasing the share of monetary activity that is productive rather than extractive, and improving population survival.* The end of the billionaire class is the visible side effect. The proposal is about ownership.

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A Cap-value sweep data

Table 3: Full results of the cap-value sweep at the joint-calibrated UK baseline. $N = 10{,}000$, 4 seeds, 180 ticks. Solid line in Figure 5.

arm	cap	alive %	≥£1B	≥£100M	≥£10M	top-1%	Pareto slope
cap alone	£1M	80%	0	27	996	0.189	−0.698
cap alone	£5M	83%	0	208	1,424	0.198	−0.455
cap alone	£10M	83%	6	403	4,016	0.222	−0.454
cap alone	£50M	83%	86	721	4,084	0.323	−0.443
cap alone	£100M	83%	318	840	4,084	0.352	−0.442
cap alone	£500M	83%	700	848	4,084	0.391	−0.440
cap alone	£1B	83%	803	848	4,084	0.405	−0.440
cap alone	no cap	83%	804	848	4,084	0.327	−0.260
cap + Zucman 2%	£1M	80%	0	16	980	0.146	−0.540
cap + Zucman 2%	£5M	83%	0	62	1,414	0.135	−0.333
cap + Zucman 2%	£10M	83%	0	98	3,998	0.140	−0.321
cap + Zucman 2%	£50M	83%	6	593	4,057	0.230	−0.296
cap + Zucman 2%	£100M	83%	44	804	4,057	0.272	−0.291
cap + Zucman 2%	£500M	83%	497	804	4,057	0.335	−0.282
cap + Zucman 2%	£1B	83%	800	804	4,057	0.353	−0.263
cap + Zucman 2%	no cap	83%	800	804	4,057	0.293	−0.226